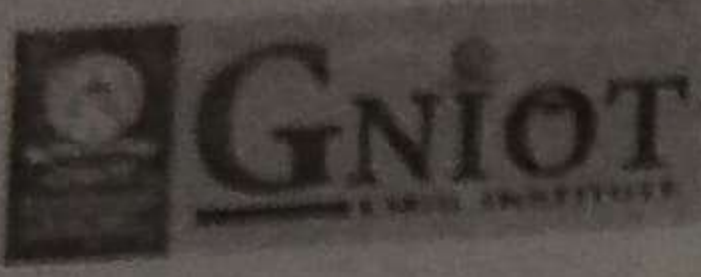


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ग्रेटर नोएडा इंस्टीट्यूट ऑफ टेक्नोलॉजी (इंजीनियरिंग इंस्टीट्यूट)
GREATER NOIDA INSTITUTE OF TECHNOLOGY (Engg. Institute)

1st CLASS TEST, SESSION 2023- 2024 (ODD SEM.)

COURSE- B.Tech

DEPTT.-CSE-IoT

YEAR:-B.TECH 2nd (3rd SEMESTER)

SUBJECT NAME:- Computer Organization and Architecture (SUBJECT CODE:- BCS-302) Set-A

MAX MARKS: 30

TIME: 2.00 Hours

Note: Answer all questions. Draw neat sketch, wherever necessary. Suitably assume missing data, if any.

SECTION A

[1X5=5]

Question 1: Attempt all parts in brief.

a.	Define computer architecture in brief.	1	[CO 1], K1, K2
b.	What is system bus?	1	[CO 1], K1, K2
c.	What is word in context to memory?	1	[CO 1], K1, K2
d.	Represent -25 and -25 in signed number representation.	1	[CO 2], K2, K4
e.	Perform 1 bit logical shift right operation on 11011101.	1	[CO 2], K2, K4

SECTION B

[3X5=15]

Question 2: Attempt any 05 out of the following.

a.	Explain register, bus and memory transfer	3	[CO 1], K1, K2
b.	Explain bus interconnection using three state buffer.	3	[CO 1], K1, K2
c.	Write short notes on processor organization.	3	[CO 1], K1, K2
d.	Differentiate polling method and independent request method of bus arbitration.	3	[CO 1], K1, K2
e.	Explain the flow chart of restoring division method.	3	[CO 2], K2, K4
f.	Multiply 7 and 5 using signed number multiplication.	3	[CO 2], K2, K4
g.	Describe 2 by 2 array multiplier with the help of neat circuit diagram.	3	[CO 2], K2, K4

SECTION C

[5X2=10]

Question 3: Attempt any 02 out of the following.

a.	Explain look ahead carry adder with help of circuit diagram.	5	[CO 1], K1, K2
b.	Multiply -6 and 5 using booth multiplication algorithm.	5	[CO 2], K2, K4
c.	Represent $(324.425)_{10}$ and $(0001011111.1101)_2$ in single precision format.	5	[CO 2], K2, K4

Bloom's Taxonomy:

K1	K2	K3	K4	K5	K6
Remember	Understand	Apply	Analyze	Evaluate	Create

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